Integrating Wound Location Data with Deep Learning for Improved Wound Classification

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Abstract—Wound classification is a crucial step in wound diagnosis. An effective classifier can aid wound specialists in identifying wound types with reduced financial and time investments, facilitating the determination of optimal treatment procedures. This study presents a deep neural network-based based classifier that leverages wound images and their corresponding locations to categorize wounds into various classes, such as diabetic, pressure, surgical, and venous ulcers. By incorporating a developed body map, the process of tagging wound locations is significantly enhanced, providing healthcare specialists with a more efficient tool for wound analysis. We conducted a comparative analysis between two prominent convolutional neural network models, ResNet50 and MobileNetV2, utilizing a dataset of 730 images. Our findings reveal that the RestNet50 outperforms MovileNetV2, achieving an accuracy of approximately 90%, compared to MobileNetV2's 83%. This disparity highlights the superior capability of ResNet50 in the context of this dataset. The results underscore the potential of integrating deep learning with spatial data to improve precision and efficiency of wound diagnosis, ultimately contributing to better patient outcomes and reducing healthcare costs.

Keywords—woundclassification,Mobilenetv2,Resnet50,Multimodel

I. Introduction

Wounds are a major problem for healthcare systems worldwide. In the United States alone, 11 million people suffer from acute wounds every year, while more than 6 million are affected by chronic wounds [1]. If not treated frequently by professional healthcare workers - the wounds are at risk for infection, amputation or even death [2].This results in a huge cost to Medicare, estimated to be between 28.1 and 96.8 billion dollars.

The types of wounds that are most common in patients include diabetic wounds, venous wounds, pressure wounds, and surgical wounds [3]. These wounds all have different yet large-scale impacts on livelihood. For example, a diabetic wound is present in about 34% of diabetic patients and almost 50% of these wounds get infected. Other significant wounds such as surgical wounds affect about 4.5% of people who have surgery every year. Not only do they serve as financial burdens but also life-threatening issues to patients. [3]

With the overwhelming amount of chronic wounds that are present in everyday life, it's a challenge to healthcare systems everywhere. Traditionally, wound evaluation has been the domain of medical specialists, especially a plastic surgeon who possesses extensive training in wound healing and management. The demand for wound assessment is growing due to the rising number of patients with conditions like diabetes and pressure injuries [4].

Currently, wound specialists manually classify wounds and document their findings in the electronic health record (EHR). However, as Artificial Intelligence (AI) further develops, these new methods are being used more as algorithms such as Machine Learning (ML) and Deep Learning (DL) are more effective and efficient specifically in wound image analysis tasks like segmentation and classification. [4]

As machine learning and deep learning continue to revolutionize the healthcare industry, the field of wound care has been significantly impacted by the integration of AI-driven tools. These advancements, particularly in the form of AI-enhanced cameras, have the potential to dramatically reduce the time and cost associated with wound diagnosis. The rapid development of deep learning techniques, supported by the availability of extensive public datasets, has led to the creation of sophisticated diagnostic tools capable of analyzing wound images with a high degree of accuracy. These tools not only enhance the precision of diagnoses but also play a crucial role in supporting clinical decision-making processes .

However, the current automation capabilities of AI models still face limitations. While algorithms for image-based wound classification have shown promise, they often overlook a critical factor: the location of the wound. This aspect, although crucial for improving classification accuracy, is frequently disregarded in existing models . Despite the importance of wound location, as commonly recorded in electronic health records (EHRs) and utilized by wound care specialists, this data is often manually documented without standardized criteria, leading to inconsistencies and reduced reliability.

As artificial intelligence continues to evolve, there is growing interest in its potential to address this gap. This paper will specifically explore the use of vision based pre-trained classification models in determining the location of wounds, with a focus on identifying and analyzing wounds on the right and left legs.

II. Related Work

The application of deep learning in wound analysis has been increasingly recognized for its potential to enhance both efficiency and accuracy in clinical decision-making. Rostami et al. [1] demonstrated the efficacy of deep convolutional neural network (DCNN)-based classifiers for wound classification, achieving accuracy rates of 91.2% for binary classifications and 82.9% for three-class classifications. These results highlight the potential of deep learning models to significantly

improve the reliability of automated wound analysis, providing valuable support in clinical applications.

Building on this foundation, Anisuzzaman et al. [3] proposed a multi-model wound classification system utilizing deep neural networks. This approach not only focused on the classification of wound images but also integrated wound location detection, which is a crucial factor often overlooked in automated systems. Their method included the development of a body map for tagging wound locations, enhancing the precision and utility of wound classification tools for healthcare professionals.

The use of advanced deep learning architectures for wound classification has also been explored by Lee et al. [4], who developed a reliable classification system using a VGG19 network. By training the model on a substantial dataset of 8,173 images and testing it on 904 images, they classified wounds into nine distinct categories. Their work underscores the potential of such networks in not only assisting healthcare professionals but also empowering patients and non-specialists in the identification and management of wounds.

Zhang et al. [5] provided a comprehensive review of existing wound image datasets and deep learning models used for wound analysis. They identified common challenges and proposed enhancements to existing models, thereby advancing the field of wound image analysis. Their insights into the development prospects of deep learning for wound care contribute to a broader understanding of how these technologies can be refined and applied effectively in clinical settings.

Abubakar et al. [6] explored the use of pre-trained deep learning algorithms, including VGG-face, ResNet101, and ResNet152, to extract features for wound classification. By employing a Support Vector Machine (SVM) for the classification task, they achieved a remarkable recognition accuracy of up to 99.9%. This high level of accuracy demonstrates the effectiveness of combining deep feature extraction with traditional machine learning classifiers in medical image analysis.

In a complementary study, Goyal et al. [7] employed a combination of traditional techniques, deep learning, and convolutional neural networks (CNNs) to classify ischemic versus non-ischemic wounds and infected versus non-infected wounds. Their approach yielded a 90% accuracy rate in ischemia classification and approximately 70% accuracy in infection classification, underscoring the utility of CNNs in the nuanced task of identifying different wound conditions.

Shenoy et al. [8] further advanced the field by developing a CNN-based method to classify 1,335 wound images into nine distinct categories, including infection, granulation tissue, and drainage, among others. Their research highlighted the capacity of CNNs to effectively analyze and classify wound images using only image pixels and corresponding labels. This study illustrates the broader trend of leveraging CNNs in medical imaging, particularly in tasks that require detailed visual analysis such as wound classification.

The early approaches to wound classification primarily relied on traditional machine learning methods, which often involved extensive feature engineering and were limited in their ability to capture complex data patterns. For instance, Rashid et al. [9] discussed the limitations of such traditional methods, emphasizing the need for more sophisticated models capable of handling the intricacies of medical image data.

Anisuzzaman et al. [10] explored the use of nine pre-trained models for classifying wound images, achieving an accuracy of 68.49% for multi-class classification and between 77% to 81% for binary classification. Their work highlights the potential of deep learning models to act as effective classifiers that facilitate quicker, more affordable wound categorization, thus supporting healthcare professionals in selecting optimal treatment strategies.

Alzubaidi et al. [11] expanded on the application of deep CNNs to the classification of Diabetic Foot Ulcers (DFUs), a significant complication of diabetes. Using a dataset of 754 images, the researchers trained a CNN to differentiate between healthy and ulcerated skin, demonstrating the efficacy of deep learning in addressing specific wound types associated with chronic conditions.

Esteva et al. [12] conducted a landmark study where a CNN was trained on 129,450 clinical images covering 2,032 different diseases. The CNN achieved performance levels comparable to those of dermatologists in skin cancer classification. This study exemplifies how deep neural networks, when combined with the ubiquity of mobile devices, have the potential to extend clinical expertise beyond traditional settings, particularly in dermatological diagnostics.

In related research, Nelson et al. [13] focused on leg ulcers, evaluating 13 interventions to determine their efficacy and safety in providing health advice to patients. Their work provides crucial insights into the treatment of leg ulcers, a common and complex wound type, by systematically categorizing various therapeutic approaches.

Coetzee et al. [14] contributed to the understanding of body mapping as a research tool, exploring its application in social and behavioral studies. The paper discusses how body mapping, when used alongside other methodologies, can offer valuable insights, particularly in understanding wound location, which is critical for accurate diagnosis and treatment planning.

The evolution of deep learning techniques, as seen in the pioneering work by Krizhevsky et al. [15], has laid the foundation for modern CNN applications in wound analysis. They trained a large CNN on 1.2 million images, achieving a top-1 error rate of 37.5% and a top-5 error rate of 17.0%, significantly outperforming previous methods. Their work demonstrated the capability of CNNs to produce exceptional results on challenging datasets, underscoring the potential of deep learning in complex image recognition tasks.

Building on this, Simonyan and Zisserman [16] examined the impact of network depth on CNN accuracy in large-scale image recognition. They introduced two high-performing ConvNet models that have since become benchmarks in the field, enabling other researchers to apply these advancements to various image classification challenges, including wound analysis.

III. Materials and Methods

The following sections of this paper will detail the methodology employed in our study, starting with an overview of the dataset used, followed by the steps taken to process the data. We will then delve into the specific techniques and models utilized, including the application of convolutional neural networks (CNNs) such as ResNet50 and MobileNetV2. These models were fine-tuned and evaluated using various metrics to determine their efficacy in wound classification. Through this structured approach, the paper aims to demonstrate the effectiveness of integrating image and location-based features for enhancing wound classification accuracy.

A. Dataset

Over a two-year long clinical period, the AZH dataset was collected at the AZH Wound and Vascular Center in Milwaukee [3]. The dataset consists of 730 wound images of varying heights and widths. The dataset consisted of five categories of wounds: diabetic, pressure, surgical, normal, or venous. The different wounds were captured by two different cameras - an iPad Pro and a Canon SX 620 HS digital camera.

Examples of these pictures are seen in Figure 2. The labeling was done by a wound specialist at the AZH Wound and Vascular Center. Each wound is unique, whether that be that they are from different patients or if they're from the same patient - they are different wounds so they're considered unique. The wound classification process is displayed visually in Figure 1.

Beginning with using this dataset as a source, the data processing began by using the body map with wound locations to generate labels for the training and testing CSV files. Integrating a detailed body map can enhance the accuracy of wound detection and classification by providing additional location-based features. By using this detailed map, it provides the model a more holistic set of data for classification. The body map allows the classifier to be trained on two aspects - image and location features - which makes it outperform classifiers that rely just on image data [17]. Following that, both training and testing files were concatenated and then the data was split into 10% for the validation, 80% for the training, and 10% for the testing dataset. After splitting the data, we used a convolution model to train both the training and validation dataset.

B. Methodology

For our experiments, we utilized two prominent convolutional neural networks, ResNet50 and MobileNet, both available in the Keras Applications library. We started by selecting ResNet50 and MobileNet as our base models, each pretrained on the ImageNet dataset. These models are known for their robust feature extraction capabilities.

To leverage the learned features from the original dataset, we froze the layers of both ResNet50 and MobileNet. This means that the weights of these layers were not updated during the training process, allowing the models to act as fixed feature extractors. The hyper parameters that were fine tuned using these models were the learning rates, which ranged from 0.000001 to 0.05, and the epochs, which ranged from 10 to 50, in intervals of 10.

On top of the frozen base models, we added several new trainable layers. Specifically, after obtaining the feature outputs from each model, we concatenated these features and then passed them through a series of dense layers. These dense layers were designed to learn and adapt to the new dataset's characteristics.The final layer in our architecture was a SoftMax layer, which provided the probability distribution over the output classes. This setup enabled us to classify the inputs into the desired categories based on the new dataset.This approach allowed the model to fine-tune its understanding of the new data without overfitting to the limited size of the dataset.

Finally, the best model is saved. The best model is evaluated using the testing dataset using metrics such as accuracy, precision, recall, and F1-score, and comparing the results across the different architectures. The evaluation reports consist of but are not limited to confusion matrices, precision and recall, and accuracy graphs.

IV. Results

In this experiment, we identified the optimal learning rate for training MobileNetV2 by evaluating its impact on validation performance across various epochs.

The results highlight that a learning rate of 0.0001 demonstrates high consistency in validation accuracy, starting at 80 % for 10 epochs and maintaining an average accuracy of approximately 83 % to 86% for Left leg. This balance is crucial for ensuring the model's robustness and generalization capabilities, particularly for specific anatomical regions like the left leg, which may present unique challenges in terms of wound variability and complexity. Similarly for the right leg, the results displayed a validation accuracy between 73 % and 77 % at a learning rate of 0.005 beginning with 10 epochs.

In Figure 3, both the right leg and the left leg's confusion matrices are displayed. These confusion matrices show how many of the wounds were accurately predicted by AI and if they were not - which wound they were actually predicted as. For example, in the right leg confusion matrix 4 of the 8 diabetic wounds were accurately predicted as diabetic wounds while 2 of the 8 diabetic wounds were predicted as surgical wounds and 1 of the 8, a venous wound. Both confusion matrices yield a relatively high accuracy rate. Below, in Figure 5, the accuracy graphs for both the right and left leg are shown. As shown in the graphs, both the accuracy values are pretty high for both the right and left leg around learning rates of 0.001 to 0.005. This plot displays that the model is pretty accurate when making predictions on training data throughout the training process. Along with that, we have the precision recall tables as well. Precision measures the accuracy of positive predictions, while recall measures their completeness. The left values are significantly higher than the right - which displays more accuracy in predictions in the left leg as compared to the right leg.

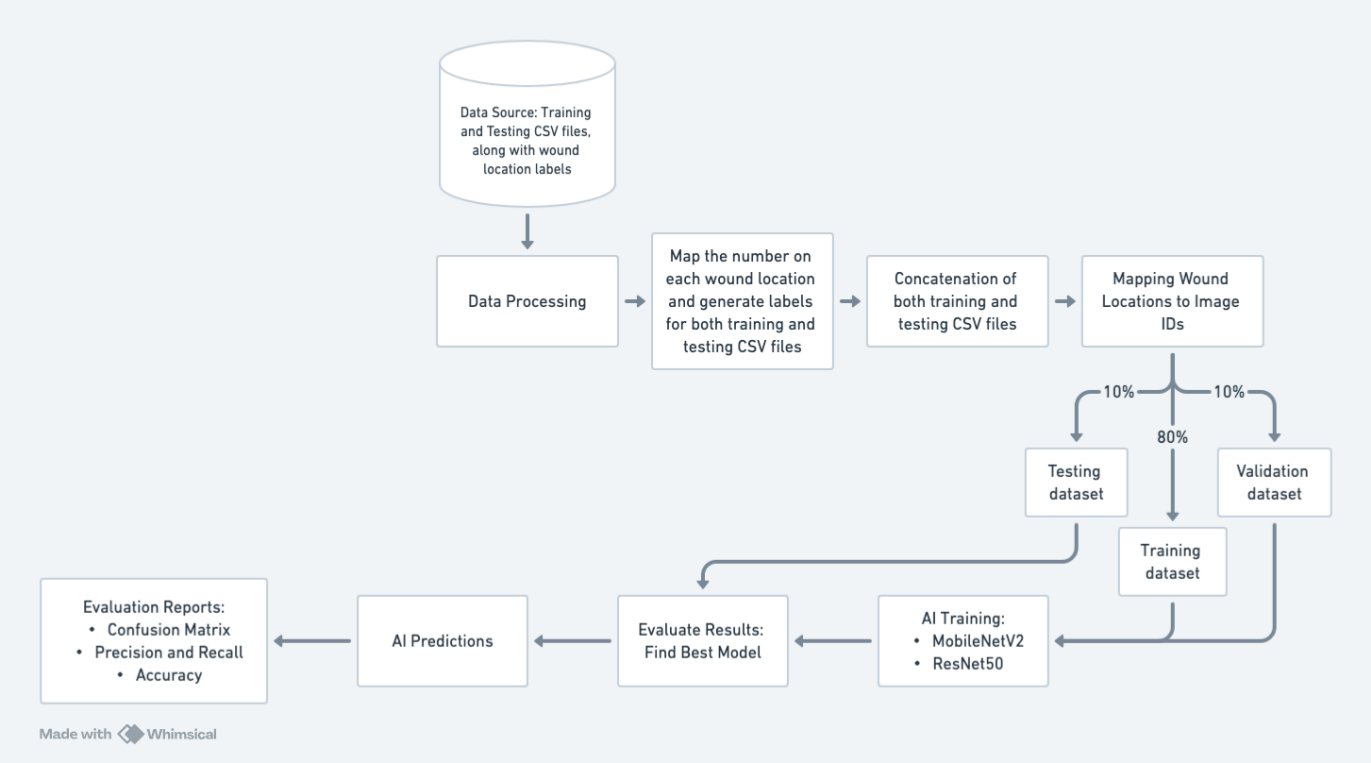


Fig. 1: Flow Chart of Wound Classification Process

(a) Diabetic Wound

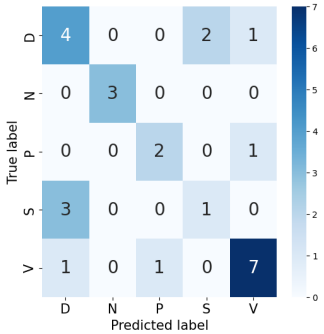
(b) Venous Wound

(c) Surgical Wound

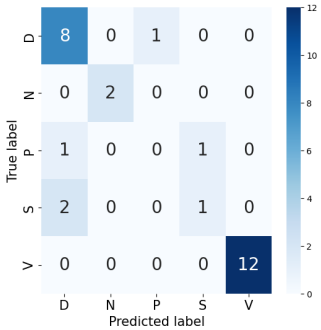
(d) Pressure Wound

Fig. 2: Images from the dataset

In addition, we also identified the optimal learning rate for training ResNet50 using different epochs to find its impact on validation performance. For the left leg, results depict that at a learning rate of 0.0001 there's about an 90% validation accuracy starting at 10 epochs. Similarly, for the right leg,



(a) Right leg

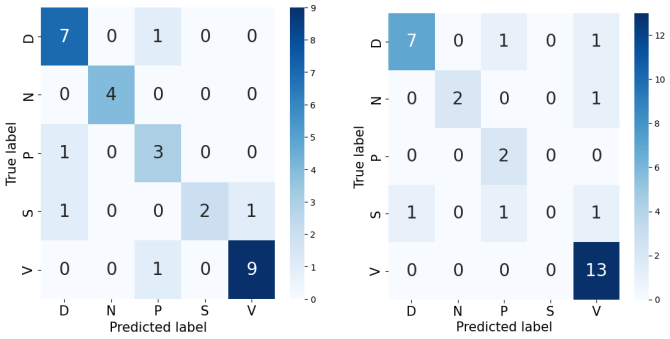


(b) Left Leg

Fig. 3: Confusion Matrix - MobileNet

there's about an 80% validation accuracy starting at 20 epochs for a learning rate of 0.0001. Both results display a high validation accuracy ensuring that the model is providing accurate predictions. In Figure 4, similar to Figure 3, the confusion matrices are shown - most of the AI predictions are accurate and classifying the wounds correctly. In addition, in Figure 6, there are accuracy graphs as well in which for the right leg at a learning rate of 0.0001 there's a pretty high accuracy of 0.8 and for the left leg at a learning rate of 0.0001 and 0.005 there's a pretty high accuracy of 0.9.

In summary, the study illustrates that while both MobileNetV2 and ResNet50 can be effective in wound classification, ResNet50 offers better accuracy and stability across



(a) Right leg (b) Left Leg

Fig. 4: Confusion Matrix - ResNet

TABLE I: Classification Report for Algorithms

Algorthim	Wound Location	Recall	Precision	F1-Score
MobilenetV2	Left Leg	0.89	0.73	0.80
MobilenetV2	Right leg	1.00	0.27	0.42
Resnet50	Left Leg	1.00	0.75	0.86
Resnet50	Right Leg	0.43	0.38	0.40

different conditions. The findings suggest that fine-tuning hyperparameters, particularly the learning rate, is crucial for optimizing model performance. Future work should explore additional strategies, such as data augmentation and the inclusion of more diverse datasets, to further enhance the generalization capabilities of these models in real-world clinical settings.

V. Conclusion

This study has demonstrated the efficacy of integrating wound location data with advanced deep learning techniques to improve wound classification accuracy. Utilizing a dataset comprising 730 wound images, we employed two well-established convolutional neural networks, ResNet50 and MobileNetV2, to classify wounds into distinct categories, including diabetic, pressure, surgical, and venous ulcers. Our experimental results reveal that ResNet50 outperforms MobileNetV2, achieving a higher accuracy rate of approximately 90%, compared to MobileNetV2's 83%. This performance disparity underscores the enhanced capability of ResNet50 in handling complex medical image classification tasks.

The incorporation of a detailed body map, which provided spatial context to wound images, proved to be a significant factor in enhancing classification performance. This method allowed for the dual consideration of image features and wound location data, which together contributed to more accurate and reliable classifications. By reducing the time and financial costs associated with wound diagnosis, this approach supports improved clinical decision-making, ultimately leading to better patient outcomes.

In our research, we had limitations of data scarcity. With more data, the model would be exposed to better training and if researched further with a larger amount of data - the model will work better for more situations. The use of this model in medical fields will allow for patients to identify wounds themselves - leading to higher efficiency and lower cost. Results that are easier to interpret can help clinicians have different and useful perspectives during the diagnostic phase [18]. Right now, chronic wounds affect 4.5 million individuals in the US annually costing almost 25 billion dollars per year [19]. Even with issues in perspective distortions, patients can have significantly varying examinations, which is also being researched to be solved by technology [20]. With technological solutions, especially for rural areas, where resources are less accessible, patients will have more accurate results and proper treatment for a lower cost. These proper treatments will be able to be found quicker and easier with technology allowing for higher efficiency and lower costs to improve patient outcomes. By having technology readily accessible to wound care providers, the burden will be reduced.

These findings suggest that deep learning models, when augmented with spatial data, hold significant potential for advancing medical image analysis. Future work should explore the integration of additional modalities and the application of even more sophisticated models to further refine and expand the capabilities of AI-driven diagnostic tools in clinical settings. In addition, with more diverse datasets, models can be trained to easily identify over a larger set of wound types.

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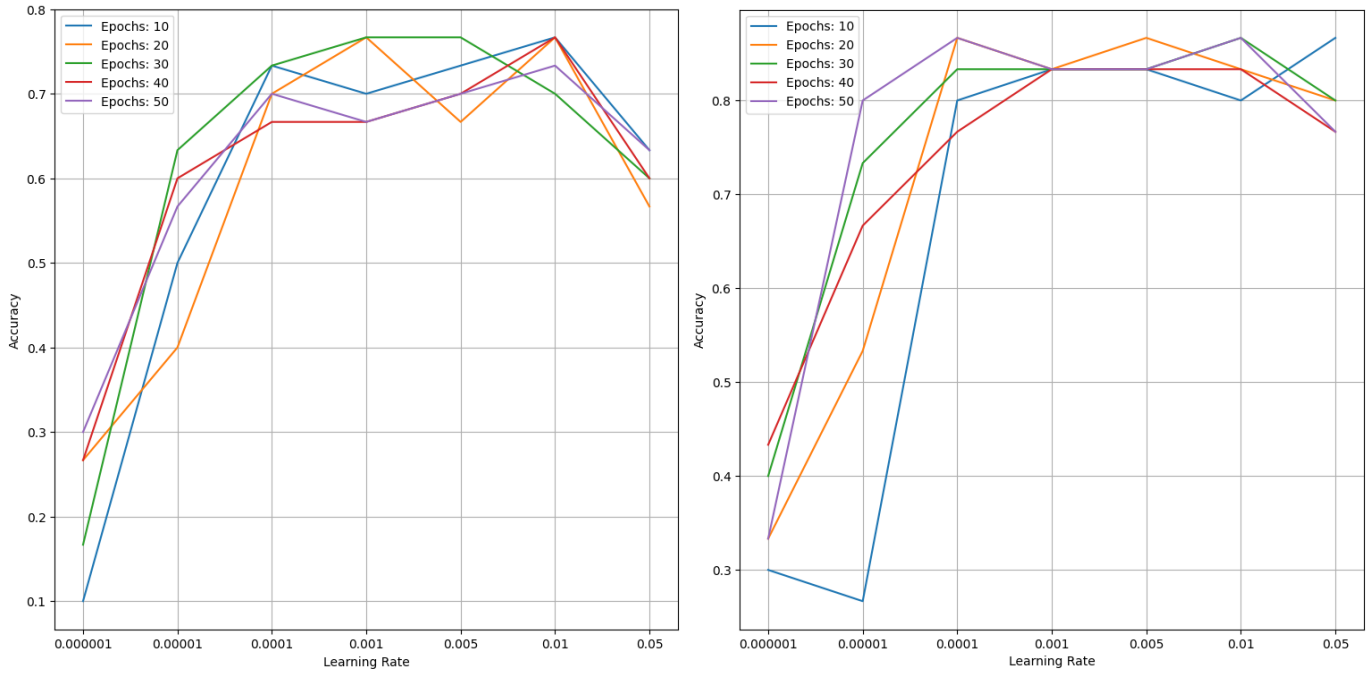
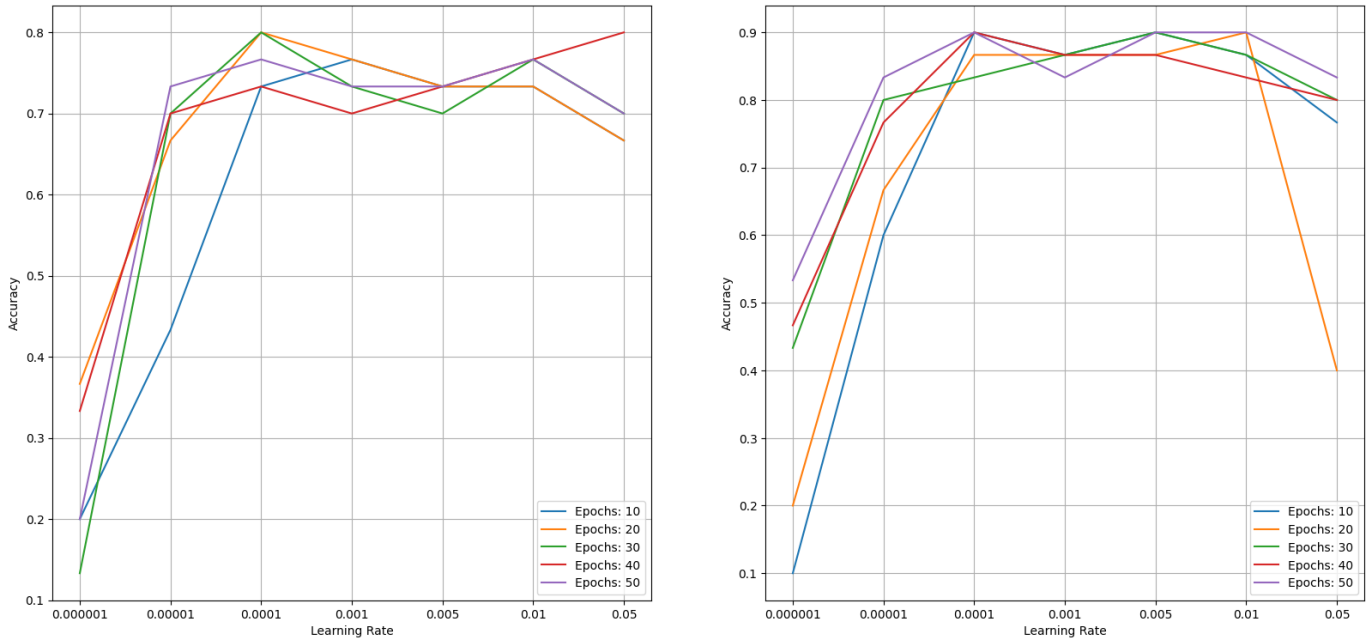


(a) Right leg (b) Left leg

Fig. 5: Accuracy Graph - MobileNet



(a) Right leg (b) Left Leg

Fig. 6: Accuracy Graph - ResNet

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